

REMARK. Notice that $\mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} = 0$ iff $\mu_i g_i(\mathbf{x}^*) = 0$ for all i .

Theorem 4.6. (*Second-Order Necessary Conditions*)

Let $f, \mathbf{g}, \mathbf{h} \in C^2$ and let $\mathbf{x}^*$ be a regular point of constraints $\mathbf{h}(\mathbf{x}) = 0$, $\mathbf{g}(\mathbf{x}) \leq 0$ and a local minimum point of f subject to these constraints. Then there exist $\boldsymbol{\lambda} \in \mathbb{R}^m$ and $\boldsymbol{\mu} \in \mathbb{R}^q$ with $\mu \geq 0$ such that

$$\begin{aligned}\nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*)\boldsymbol{\lambda} + \nabla \mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} &= 0 \\ \mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} &= 0.\end{aligned}$$

Moreover, $L(\mathbf{x}^*) = \nabla^2 f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla^2 h_i(\mathbf{x}^*) + \sum_{j=1}^q \mu_j \nabla^2 g_j(\mathbf{x}^*)$ is positive semi-definite on the tangent subspace of the active constraints at $\mathbf{x}^*$.

Proof. Since $\mathbf{x}^*$ is a local minimum point over the constraints $\mathbf{h}(\mathbf{x}) = 0$ and $\mathbf{g}(\mathbf{x}) \leq 0$, it is also a local minimum point for the problem with the active constraints taken as equality constraints. So the original problem can be replaced by the corresponding problem having equality constraints only. $\square$

Theorem 4.7. (*Second-Order Sufficient Conditions*)

Let $f, \mathbf{g}, \mathbf{h} \in C^2$ and let $\mathbf{x}^* \in \mathbb{R}^n, \boldsymbol{\lambda} \in \mathbb{R}^m, \boldsymbol{\mu} \in \mathbb{R}^q$ be such that

- (i) $\mathbf{h}(\mathbf{x}^*) = 0, \quad \mathbf{g}(\mathbf{x}^*) \leq 0;$
- (ii) $\boldsymbol{\mu} \geq 0;$
- (iii) $\mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} = 0;$
- (iv) $\nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*)\boldsymbol{\lambda} + \nabla \mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} = 0;$
- (v) $L(\mathbf{x}^*) = \nabla^2 f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla^2 h_i(\mathbf{x}^*) + \sum_{j=1}^q \mu_j \nabla^2 g_j(\mathbf{x}^*)$ is positive definite on the subspace

$$M = \{\mathbf{y} : \nabla \mathbf{h}(\mathbf{x}^*)^T \mathbf{y} = 0, \nabla g_j(\mathbf{x}^*)^T \mathbf{y} = 0 \quad \forall j \in J\}$$

where $J = \{j : g_j(\mathbf{x}^*) = 0, \mu_j > 0\}$.

Then $\mathbf{x}^*$ is a local minimum point of $f(\mathbf{x})$ subject to $\mathbf{h}(\mathbf{x}) = 0$ and $\mathbf{g}(\mathbf{x}) \leq 0$. $\square$

§5. Lagrange Methods

Lagrange methods are solution methods for constrained optimization problems based on directly solving the Lagrange first-order necessary conditions. For the equality constrained problem

$$\begin{array}{ll} \text{Min} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{h}(\mathbf{x}) = 0 \end{array}$$

where $\mathbf{x}$ is n -dimensional and $\mathbf{h}(\mathbf{x})$ is m -dimensional, this approach amounts to solving the system of equations

$$\begin{cases} \nabla f(\mathbf{x}) + \nabla \mathbf{h}(\mathbf{x})\boldsymbol{\lambda} = 0 \\ \mathbf{h}(\mathbf{x}) = 0 \end{cases}$$

for $\mathbf{x}$ and $\boldsymbol{\lambda}$. The set of necessary conditions is a system of $n + m$ equations in the $n + m$ unknowns comprising the components of $\mathbf{x}$ and $\boldsymbol{\lambda}$.

EXAMPLE 5.1. Let A be a symmetric matrix. Solve

$$\begin{array}{ll} \text{Min} & \mathbf{x}^T A \mathbf{x} \\ \text{s.t.} & \mathbf{x}^T \mathbf{x} = 1. \end{array}$$

SOLUTION. Set $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ and $h(\mathbf{x}) = \mathbf{x}^T \mathbf{x} - 1$. By the first-order necessary condition, we have

$$\begin{cases} \nabla f(\mathbf{x}) + \nabla h(\mathbf{x})\lambda = 0 \\ h(\mathbf{x}) = 0. \end{cases}$$

Or

$$\begin{cases} 2A\mathbf{x} + 2\lambda\mathbf{x} = 0 \\ \mathbf{x}^T \mathbf{x} = 1. \end{cases}$$

Let μ stand for $-\lambda$. Then

$$\begin{cases} A\mathbf{x} = \mu\mathbf{x} & (5.1) \\ \mathbf{x}^T \mathbf{x} = 1. & (5.2) \end{cases}$$

Hence, by (5.1) and (5.2), $\mathbf{x}^T A \mathbf{x} = \mu \mathbf{x}^T \mathbf{x} = \mu$. It follows from (5.1) that μ is an eigenvalue of A and $\mathbf{x}$ is the corresponding eigenvector. Since the feasible region is closed and bounded, by a theorem in Calculus, f achieves minimum at some feasible point. Hence the minimum value of f is the smallest eigenvalue of A which occurs at a corresponding eigenvector with unit norm. $\square$

EXAMPLE 5.2. Let $Q \in \mathbb{R}^{n \times n}$ be a positive definite matrix and let $A \in \mathbb{R}^{m \times n}$ be of rank m . Solve

$$\begin{array}{ll} \text{Min} & \frac{1}{2} \mathbf{x}^T Q \mathbf{x} + \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & A\mathbf{x} = \mathbf{b}. \end{array}$$

SOLUTION. The Lagrange necessary conditions for this problem are

$$\begin{cases} Q\mathbf{x} + A^T \boldsymbol{\lambda} + \mathbf{c} = 0 \\ A\mathbf{x} - \mathbf{b} = 0. \end{cases}$$

Thus

$$\begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \lambda \end{bmatrix} = \begin{bmatrix} -\mathbf{c} \\ \mathbf{b} \end{bmatrix}. \quad (5.3)$$

Since

$$\begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} Q^{-1} & -Q^{-1}A^T \\ 0 & I \end{bmatrix} = \begin{bmatrix} I & 0 \\ * & -AQ^{-1}A^T \end{bmatrix} \quad (5.4)$$

and since both $\begin{bmatrix} Q^{-1} & -Q^{-1}A^T \\ 0 & I \end{bmatrix}$ and $\begin{bmatrix} I & 0 \\ * & -AQ^{-1}A^T \end{bmatrix}$ are nonsingular, from (5.4) it can be seen that $\begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix}$ is nonsingular. In view of (5.3), we have

$$\begin{bmatrix} \mathbf{x} \\ \lambda \end{bmatrix} = \begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix}^{-1} \begin{bmatrix} -\mathbf{c} \\ \mathbf{b} \end{bmatrix}. \quad (5.5)$$

$\mathbf{x}^*$ is the first entry of the vector on the RHS of (5.5).

Note that $L(\mathbf{x}^*) = \nabla^2 f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla^2 h_i(\mathbf{x}^*) = Q$, which is positive definite. By the second-order sufficient condition, $\mathbf{x}^*$ is a local minimum point. Since Q is positive definite, $f(\mathbf{x})$ is a convex function by Theorem 1.11. Note that the feasible region is a convex set, $\mathbf{x}^*$ is a global minimum point by Theorem 1.13. $\square$

Now let us apply the Lagrange methods to constrained optimization problems involving both equality and inequality constraints. We illustrate these methods through one concrete example.

EXAMPLE 5.3. Consider the problem

$$\begin{array}{ll} \text{Min} & 2x_1^2 + 2x_1x_2 + x_2^2 - 10x_1 - 10x_2 \\ \text{s.t.} & x_1^2 + x_2^2 \leq 5 \\ & 3x_1 + x_2 \leq 6 \end{array}$$

SOLUTION. The first-order necessary conditions, in addition to the constraints, are

$$\begin{cases} 4x_1 + 2x_2 - 10 + 2\mu_1x_1 + 3\mu_2 = 0 \\ 2x_1 + 2x_2 - 10 + 2\mu_1x_2 + \mu_2 = 0 \\ \mu_1 \geq 0, \quad \mu_2 \geq 0 \\ \mu_1(x_1^2 + x_2^2 - 5) = 0 \\ \mu_2(3x_1 + x_2 - 6) = 0. \end{cases}$$

To find a solution, we define various combinations of active constraints, check the signs of the resulting μ_1 and μ_2 , and check feasibility of the resulting $\mathbf{x}^*$. In this problem we need to distinguish among three cases, that is, we should try setting none, one, or two constraints active. We only consider one case here and the details of other two cases are omitted.

Assume that the first constraint is active and the second is inactive, we have the equations

$$\begin{cases} 4x_1 + 2x_2 - 10 + 2\mu_1x_1 = 0 \\ 2x_1 + 2x_2 - 10 + 2\mu_1x_2 = 0 \\ x_1^2 + x_2^2 = 5, \end{cases}$$

which yields $x_1 = 1$, $x_2 = 2$ and $\mu_1 = 1$. Thus $3x_1 + x_2 = 5$ and hence the second constraint is satisfied.

Note that

$$\begin{aligned} L(\mathbf{x}^*) &= \nabla^2 f(\mathbf{x}^*) + \sum \mu_i \nabla^2 g_i(\mathbf{x}^*) \\ &= \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}, \end{aligned}$$

which is positive definite. By the second-order sufficient conditions, $\mathbf{x}^*$ is a local minimum point of f . Since f is a convex function and the feasible region is a convex set, $\mathbf{x}^*$ is a global minimum point. □

EXERCISE 5.1. Solve Example 1.4 by a Lagrange method.

§6. Penalty Methods

Penalty methods are procedures for approximating constrained optimization problems by unconstrained problems. The approximation is accomplished by adding to the objective function a term that prescribes a high cost for violation of the constraints.

Consider the problem

$$\begin{array}{ll} \text{Min} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in \Omega \end{array} \quad (6.1)$$

where f is continuous on $\mathbb{R}^n$, Ω is the feasible region which is usually defined by a number of functional constraints.

Replace (6.1) by an unconstrained problem of the form

$$\text{Min } f(\mathbf{x}) + cp(\mathbf{x}) \quad (6.2)$$

where c is a positive constant and $p(\mathbf{x})$ is a function defined on $\mathbb{R}^n$ satisfying

- (i) $p(\mathbf{x})$ is continuous;
- (ii) $p(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$;
- (iii) $p(\mathbf{x}) = 0$ iff $\mathbf{x} \in \Omega$.

Usually (6.2) is called a penalty problem, $p(\mathbf{x})$ is called a penalty function, and c is called a penalty factor or penalty coefficient.

EXAMPLE 6.1. Let $\Omega = \{\mathbf{x} : \mathbf{h}(\mathbf{x}) = 0 \text{ and } \mathbf{g}(\mathbf{x}) \leq 0\}$, where $\mathbf{h}$ is m -dimensional and $\mathbf{g}$ is q -dimensional. Then a very useful penalty function is

$$p(\mathbf{x}) = \frac{1}{2} \sum_{i=1}^m h_i(\mathbf{x})^2 + \frac{1}{2} \sum_{j=1}^q (\max\{0, g_j(\mathbf{x})\})^2$$

Idea: For large c , the minimum point of problem (6.2) will be in a region where $p(\mathbf{x})$ is small. Thus, for increasing c it is expected that the corresponding solution points will approach the feasible region Ω and, subject to being close, will minimize f . Ideally, as $c \rightarrow \infty$ the solution point of the penalty problem will converge to a solution of the constrained problem.

Method: Let $\{c_k\}$, $k = 1, 2, \dots$, be a sequence tending to infinity such that $c_k \geq 0$, $c_{k+1} > c_k$. Define

$$q(c, \mathbf{x}) = f(\mathbf{x}) + cp(\mathbf{x}).$$

For each k solve the problem

$$\text{Min } q(c_k, \mathbf{x})$$

obtaining a solution of $\mathbf{x}_k$.

Theorem 6.1. Let $\{\mathbf{x}_k\}$ be a sequence generated by the penalty method. Then any cluster (or limit) point of the sequence is an optimal solution to (6.1).

EXAMPLE 6.2. Solve the following problem by a penalty method.

$$\begin{aligned} \text{Min } & f(\mathbf{x}) = x_1 + x_2 \\ \text{s.t. } & g_1(\mathbf{x}) = x_1^2 - x_2 \leq 0 \\ & g_2(\mathbf{x}) = -x_1 \leq 0 \end{aligned}$$

SOLUTION. Consider the unconstrained problem

$$\text{Min } \phi(\mathbf{x}, c) = x_1 + x_2 + c[\max\{0, x_1^2 - x_2\}]^2 + c[\max\{0, -x_1\}]^2. \quad (6.3)$$

It is easy to see that ϕ is a convex function provided $c > 0$. Now let $q_1(\mathbf{x}) = \max\{0, x_1^2 - x_2\}$ and $q_2(\mathbf{x}) = \max\{0, -x_1\}$. Then

$$\frac{\partial q_1}{\partial x_1} = \begin{cases} 2x_1 & \text{if } x_1^2 - x_2 \geq 0, \\ 0 & \text{otherwise.} \end{cases} \quad \frac{\partial q_1}{\partial x_2} = \begin{cases} -1 & \text{if } x_1^2 - x_2 \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

$$\frac{\partial q_2}{\partial x_1} = \begin{cases} 0 & \text{if } x_1 \geq 0, \\ -1 & \text{otherwise.} \end{cases} \quad \frac{\partial q_2}{\partial x_2} = 0.$$

Hence

$$\frac{\partial \phi}{\partial x_1} = 1 + 2c \cdot 2x_1 \cdot \text{Max}\{0, x_1^2 - x_2\} - 2c \cdot \text{Max}\{0, -x_1\}. \quad (6.4)$$

$$\frac{\partial \phi}{\partial x_2} = 1 + 2c \cdot (-1) \cdot \text{Max}\{0, x_1^2 - x_2\}. \quad (6.5)$$

Setting $\frac{\partial \phi}{\partial x_1} = \frac{\partial \phi}{\partial x_2} = 0$ yields $x_1^2 - x_2 > 0$ by (6.5) and then $x_1 < 0$ by (6.4). So (6.4) and (6.5) amount to that

$$\begin{cases} 1 + 2c \cdot 2x_1(x_1^2 - x_2) + 2cx_1 = 0, \\ 1 - 2c(x_1^2 - x_2) = 0. \end{cases}$$

Thus

$$\begin{cases} x_1 = -\frac{1}{2(1+c)}, \\ x_2 = \frac{1}{4(1+c)^2} - \frac{1}{2c}, \end{cases}$$

or

$$\mathbf{x} = \left(-\frac{1}{2(1+c)}, \quad \frac{1}{4(1+c)^2} - \frac{1}{2c} \right)^T,$$

which is an optimal solution to (6.3) since ϕ is convex. Letting $c \rightarrow \infty$ gives an optimal solution $\mathbf{x}^* = (0, 0)^T$ to the original problem. $\square$

Penalty Functions

Consider problems of the form

$$\begin{array}{ll} \text{Min} & f(\mathbf{x}) \\ \text{s.t.} & h_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, m. \\ & g_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, q. \end{array}$$

Define $g_j^+(\mathbf{x}) = \text{Max} \{0, g_j(\mathbf{x})\}$ for $j = 1, 2, \dots, q$. Then the simplest and most frequently used penalty function is of the form

$$p(\mathbf{x}) = \sum_{i=1}^m |h_i(\mathbf{x})|^\alpha + \sum_{j=1}^q |g_j^+(\mathbf{x})|^\beta,$$

where α and β are even numbers.

§7. Barrier Methods

The basic motivation behind barrier methods is also to approximate constrained optimization problem by unconstrained problem. Unlike the penalty methods, the approximation is accomplished by adding a term that favors points interior to the feasible region over those near the boundary.

Barrier methods are applicable to the problems of the form

$$\begin{array}{ll} \text{Min} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in \Omega \end{array} \quad (7.1)$$

where Ω has a nonempty interior that is arbitrarily close to any point of Ω . Intuitively, this means that the set has an interior and it is possible to get to any boundary point by approaching it from the interior. We shall refer to such a set as robust. Usually, Ω takes the form

$$\Omega = \{\mathbf{x} : g_i(\mathbf{x}) \leq 0, i = 1, 2, \dots, p\}.$$

A barrier function is a function B defined on the interior of Ω such that

- (i) $B(\mathbf{x})$ is continuous;
- (ii) $B(\mathbf{x}) \geq 0$;
- (iii) $B(\mathbf{x}) \rightarrow \infty$ as $\mathbf{x}$ approaches the boundary of Ω .

Method: Let $\{c_k\}_{k=1}^{\infty}$ be a sequence tending to infinity such that $c_k \geq 0$ and $c_{k+1} > c_k$. Define the function

$$r(c, \mathbf{x}) = f(\mathbf{x}) + \frac{1}{c} B(\mathbf{x}).$$

For each k solve the problem

$$\begin{array}{ll} \text{Min} & r(c_k, \mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in \text{interior of } \Omega \end{array} \quad (7.2)$$

obtaining the point $\mathbf{x}_k$.

Theorem 7.1. *Any limit (or clutter) point of the sequence $\{\mathbf{x}_k\}$ generated by the barrier method is a solution to problem (7.1).*

REMARK. (7.2) is called the barrier problem. Despite its constraint, it can be solved using an unconstrained search technique. To find the solution one starts at an initial interior point and then searches from the point using the steepest descent method or some other methods. Since the value of the objective function approaches infinity near the boundary of Ω , the search technique will automatically remain within the interior of Ω , and the constraints need not be accounted for explicitly. Thus, although (7.2) is a constrained problem, from a computational viewpoint it is unconstrained.

Barrier functions

Consider problems of the form

$$\begin{array}{ll} \text{Min} & f(\mathbf{x}) \\ \text{s.t.} & g_i(\mathbf{x}) \leq 0, \quad i = 1, 2, \dots, p. \end{array}$$

There are two most frequently used barrier functions

1. $B(\mathbf{x}) = \sum_{i=1}^p -\frac{1}{g_i(\mathbf{x})}$;
2. $B(\mathbf{x}) = \sum_{i=1}^p -\log(-g_i(\mathbf{x}))$.

EXAMPLE 7.1. Solve the problem by a barrier method.

$$\begin{array}{ll} \text{Min} & f(\mathbf{x}) = \frac{1}{3}(x_1 + 1)^3 + x_2 \\ \text{s.t.} & g_1(\mathbf{x}) = -x_1 + 1 \leq 0 \\ & g_2(\mathbf{x}) = -x_2 \leq 0. \end{array}$$

SOLUTION. Consider the following problem

$$\text{Min } \phi(\mathbf{x}, c) = \frac{1}{3}(x_1 + 1)^3 + x_2 + \frac{1}{c} \frac{1}{x_1 - 1} + \frac{1}{c} \frac{1}{x_2} \quad (7.3)$$

subject to the constraint that $\mathbf{x}$ is an interior point of the original feasible region. Clearly, ϕ is a convex function on the interior of the feasible region.

Setting $\frac{\partial \phi}{\partial x_1} = \frac{\partial \phi}{\partial x_2} = 0$ gives

$$\begin{cases} (x_1 + 1)^2 - \frac{1}{c(x_1 - 1)^2} = 0, \\ 1 - \frac{1}{cx_2^2} = 0. \end{cases}$$

So we have

$$\begin{cases} x_1 = \sqrt{1 + \sqrt{\frac{1}{c}}}, \\ x_2 = \sqrt{\frac{1}{c}}, \end{cases}$$

which, in view of the convexity of ϕ , is an optimal solution of (7.3). Hence

$$\mathbf{x}^* = \lim_{c \rightarrow \infty} \left(\sqrt{1 + \sqrt{\frac{1}{c}}}, \sqrt{\frac{1}{c}} \right)^T = (1, 0)^T$$

is an optimal solution to the original problem. □

EXAMPLE 7.2. Solve the following problem by a barrier method.

$$\begin{array}{ll} \text{Min} & f(\mathbf{x}) = x_1 + x_2 \\ \text{s.t.} & g_1(\mathbf{x}) = x_1^2 - x_2 \leq 0 \\ & g_2(\mathbf{x}) = -x_1 \leq 0. \end{array}$$

SOLUTION. Consider the following problem

$$\text{Min } \phi(\mathbf{x}, c) = x_1 + x_2 - \frac{1}{c} \log(x_2 - x_1^2) - \frac{1}{c} \log x_1$$

subject to the constraint that $\mathbf{x}$ is an interior point of the original constraint set. Note that ϕ is a convex function on the interior of the feasible region.

Set $\frac{\partial \phi}{\partial x_1} = \frac{\partial \phi}{\partial x_2} = 0$, we have

$$\begin{cases} 1 - \frac{(-2x_1)}{c(x_2 - x_1^2)} - \frac{1}{cx_1} = 0, \\ 1 - \frac{1}{c(x_2 - x_1^2)} = 0. \end{cases}$$

Thus

$$(x_1, x_2) = \left(\frac{-c + \sqrt{c^2 + 8c}}{4c}, \frac{c + 12 - \sqrt{c^2 + 8c}}{8c} \right).$$

From the convexity of ϕ , it can be seen that the above solution is a minimum point of ϕ .

Hence

$$\mathbf{x}^* = \lim_{c \rightarrow \infty} \left(\frac{-c + \sqrt{c^2 + 8c}}{4c}, \frac{c + 12 - \sqrt{c^2 + 8c}}{8c} \right)^T = (0, 0)^T$$

is an optimal solution to the original problem. $\square$

REMARK. Note that the solution to the penalty problem is usually outside the feasible region, while the solution to the barrier problem is inside the feasible region. Thus the penalty method is also called the exterior penalty method and the barrier method is also called the interior penalty method in the literature.

§8. Duality Theory

Duality plays a central role in the development and unification of the theory of optimization.

Dual methods are based on the viewpoint that it is the Lagrange multipliers that are the fundamental unknowns associated with a constrained problem; once these multipliers are known, the determination of the solution point is simple (at least in some situation). Dual methods, therefore, do not attack the original constrained problem directly, but instead attack an alternate problem, the dual problem, whose unknowns are the Lagrange multipliers of the first problem.

8.1° Equality Constraints

Consider the following optimization problem, which is called the primal problem.

Primal Problem (P):

$$\begin{array}{ll} \text{Min} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{h}(\mathbf{x}) = 0 \end{array}$$

where $\mathbf{x} \in \mathbb{R}^n$, $f, \mathbf{h} \in \mathcal{C}^2$.

Several problems, closely related to the above primal problem, have been proposed in the literature, which are called dual problems. Among the various duality formulations, the Lagrangian duality formulation has perhaps attracted the most attention.

Lagrangian Dual Problem (D):

$$\text{Max}_{\boldsymbol{\lambda}} \phi(\boldsymbol{\lambda})$$

where $\phi(\boldsymbol{\lambda}) = \text{Min}_{\mathbf{x}} [f(\mathbf{x}) + \mathbf{h}(\mathbf{x})\boldsymbol{\lambda}]$, which is called the dual function.

Definition. Let $\mathbf{x}^*$ be a local minimum point of (P) and a regular point of the constraints $\mathbf{h}(\mathbf{x}) = 0$. Then the first-order necessary condition guarantees the existence of $\boldsymbol{\lambda}^*$ in $\mathbb{R}^m$ such that $\nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*)\boldsymbol{\lambda}^* = 0$. Recall that this $\boldsymbol{\lambda}^*$ is called the Lagrange Multiplier.

Theorem 8.1. (*Local Duality Theorem*)

Let $\mathbf{x}^*$ be a point with the following properties:

- (i) $\mathbf{x}^*$ is a local minimum point of (P) with corresponding value r^* and Lagrange multiplier $\boldsymbol{\lambda}^*$;
- (ii) $\mathbf{x}^*$ is a regular point of $\mathbf{h}(\mathbf{x}) = 0$;
- (iii) $L(\mathbf{x}^*) = \nabla^2 f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla^2 h_i(\mathbf{x}^*)$ is positive definite.

Then λ^* is a local maximum point of (D) with corresponding value r^* , and $\mathbf{x}^*$ is the point corresponding to λ^* in the definition of ϕ . $\square$

REMARK. By the second-order necessary condition, $L(\mathbf{x}^*)$ is positive semi-definite on $M = \{\mathbf{y} : \nabla h(\mathbf{x}^*)^T \mathbf{y} = 0\}$. Here we further assume that $L(\mathbf{x}^*)$ is positive definite on the whole $\mathbb{R}^n$. This assumption guarantees that the Lagrangian

$$\ell(\mathbf{x}) = f(\mathbf{x}) + \mathbf{h}(\mathbf{x})\lambda^*$$

is locally convex at $\mathbf{x}^*$.

EXAMPLE 8.1. Solve the following problem

$$\begin{array}{ll} \text{Min} & -x_1x_2 \\ \text{s.t.} & (x_1 - 3)^2 + x_2^2 = 5. \end{array}$$

SOLUTION. Let us apply both primal and dual methods to this problem, respectively.

Method 1. Primal Method

Set $f(\mathbf{x}) = -x_1x_2$ and $h(\mathbf{x}) = (x_1 - 3)^2 + x_2^2 - 5$. Then $\nabla f(\mathbf{x}) = (-x_2, -x_1)^T$ and $\nabla h(\mathbf{x}) = (2x_1 - 6, 2x_2)^T$. According to the first-order necessary condition, we have

$$\begin{cases} \nabla f(\mathbf{x}) + \nabla h(\mathbf{x})\lambda = 0 \\ h(\mathbf{x}) = 0, \end{cases}$$

or

$$\begin{cases} \begin{pmatrix} -x_2 \\ -x_1 \end{pmatrix} + \begin{pmatrix} 2x_1 - 6 \\ 2x_2 \end{pmatrix} \lambda = 0 \\ (x_1 - 3)^2 + x_2^2 = 5, \end{cases}$$

yielding $\mathbf{x}^* = (x_1, x_2)^T = (4, 2)^T$, $\lambda^* = 1$. Thus

$$\begin{aligned} L(\mathbf{x}^*) &= \nabla^2 f(\mathbf{x}^*) + \lambda^* \nabla^2 h(\mathbf{x}^*) \\ &= \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \end{aligned}$$

which is positive definite on $\mathbb{R}^n$. Hence by the second-order sufficient condition, $\mathbf{x}^*$ is a local minimum point with corresponding value of -8 . (In fact, $\mathbf{x}^*$ is a global minimum point, why?)

Method 2. Dual Method

Define

$$\phi(\lambda) = \text{Min} \{-x_1x_2 + \lambda[(x_1 - 3)^2 + x_2^2 - 5]\}. \quad (8.1)$$

It can be shown that

$$\phi(\lambda) = \begin{cases} -\frac{4\lambda(1+5\lambda^2)}{4\lambda^2-1} & \text{if } \lambda > \frac{1}{2} \\ -\infty & \text{otherwise,} \end{cases} \quad (8.2)$$

which, when $\lambda > \frac{1}{2}$, is achieved at

$$\begin{cases} x_1 = \frac{12\lambda^2}{4\lambda^2-1} \\ x_2 = \frac{6\lambda}{4\lambda^2-1}. \end{cases} \quad (8.3)$$

Assuming (8.2) and (8.3), we may proceed as follows. Since

$$\phi'(\lambda) = \frac{4(1-\lambda^2)(20\lambda^2+1)}{(4\lambda^2-1)^2} \quad \text{when } \lambda > \frac{1}{2},$$

setting $\phi'(\lambda) = 0$ yields $\lambda = 1$. Clearly, $\lambda = 1$ is a local maximum of ϕ and $\phi(1) = -8$. By (8.3), we have $\mathbf{x}^* = (4, 2)^T$. Note that $L(\mathbf{x}^*) = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ is positive definite, so $\mathbf{x}^*$ is a local minimum point of the primal problem.

The rest is to prove (8.2) and (8.3). Let $\ell(\mathbf{x}; \lambda)$ stand for the objective function of (8.1). Then the Hessian of $\ell(\mathbf{x}; \lambda)$ is

$$\nabla^2 \ell = \begin{bmatrix} 2\lambda & -1 \\ -1 & 2\lambda \end{bmatrix}$$

which is positive definite when $\lambda > \frac{1}{2}$. Hence $\ell(\mathbf{x}; \lambda)$ is a convex function for every $\lambda > \frac{1}{2}$; setting $\frac{\partial \ell}{\partial x_1} = \frac{\partial \ell}{\partial x_2} = 0$ yields (8.3) and hence the first line of the RHS of (8.2).

If $\lambda \leq \frac{1}{2}$ then, setting $x_1 = x_2$, we obtain

$$\ell(\mathbf{x}; \lambda) = (2\lambda - 1)x_1^2 - 6\lambda x_1 + 4\lambda.$$

Hence $\phi(\lambda) = \min_{x_1=x_2} \ell(\mathbf{x}; \lambda) = -\infty$. □

8.2° Inequality Constraints

Consider the following primal problem:

$$\begin{aligned} \text{(P)} \quad & \text{Min} \quad f(\mathbf{x}) \\ & \text{s.t.} \quad \mathbf{h}(\mathbf{x}) = 0 \\ & \quad \mathbf{g}(\mathbf{x}) \leq 0, \end{aligned}$$

where $\mathbf{x} \in \mathbb{R}^n$, $f, \mathbf{h}, \mathbf{g} \in \mathcal{C}^2$. The Lagrangian dual is defined as

$$\text{(D)} \quad \text{Max}_{\boldsymbol{\mu} \geq 0, \boldsymbol{\lambda}} \quad \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}),$$

where $\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \min_{\mathbf{x}} [f(\mathbf{x}) + \mathbf{h}(\mathbf{x})\boldsymbol{\lambda} + \mathbf{g}(\mathbf{x})\boldsymbol{\mu}]$ with $\boldsymbol{\mu} \geq 0$, which is called the dual function.

Theorem 8.2. (*Local Duality Theorem*)

Let $\mathbf{x}^*$ be a point with the following properties:

- (i) $\mathbf{x}^*$ is a local minimum of (P) with corresponding value r^* and Lagrange multipliers λ^* and $\mu^* \geq 0$;
- (ii) $\mathbf{x}^*$ is a regular point of the constraints in (P);
- (iii) $L(\mathbf{x}^*) = \nabla^2 f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla^2 h_i(\mathbf{x}^*) + \sum_{j=1}^q \mu_j \nabla^2 g_j(\mathbf{x}^*)$ is positive definite.

Then (D) has a local maximum with respect to $\mu \geq 0$, λ at μ^*, λ^* .

EXAMPLE 8.2. Solve the problem by a dual method.

$$\begin{array}{ll} \text{Min} & x_1^2 + x_2^2 \\ \text{s.t.} & 2x_1 + x_2 \leq -4. \end{array}$$

SOLUTION. By definition, the dual function is

$$\phi(\mu) = \text{Min}_{\mathbf{x}} \{x_1^2 + x_2^2 + \mu(2x_1 + x_2 + 4)\} \triangleq \text{Min}_{\mathbf{x}} \ell(\mathbf{x}; \mu),$$

where $\mu \geq 0$. Now setting $\frac{\partial \ell}{\partial x_1} = \frac{\partial \ell}{\partial x_2} = 0$ gives

$$\begin{cases} 2x_1 + 2\mu = 0 \\ 2x_2 + \mu = 0 \end{cases} \implies \begin{cases} x_1 = -\mu \\ x_2 = -\mu/2. \end{cases}$$

Since $\ell(\mathbf{x}; \mu)$ is a convex function for any fixed μ , we have

$$\phi(\mu) = (-\mu)^2 + (-\mu/2)^2 + \mu(-2\mu - \mu/2 + 4) = -\frac{5}{4}\mu^2 + 4\mu,$$

whose maximum in the case $\mu \geq 0$ occurs when $\frac{d\phi}{d\mu} = 0$, implying $\mu^* = \frac{8}{5}$. It follows that $\mathbf{x}^* = (-\frac{8}{5}, -\frac{4}{5})^T$ and $\phi(\mu^*) = -\frac{5}{4}(\frac{8}{5})^2 + \frac{32}{5} = \frac{16}{5}$. Since $L = \nabla^2 f + \mu^* \nabla^2 g = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ is positive definite, $\mathbf{x}^*$ is a local minimum point. (In fact, $\mathbf{x}^*$ is a global minimum point, why?) □

8.3° Convex Duality

Consider the problem

$$\begin{array}{ll} \text{(P)} & \text{Min} \quad f(\mathbf{x}) \\ & \text{s.t.} \quad \mathbf{h}(\mathbf{x}) = 0 \\ & \quad \mathbf{g}(\mathbf{x}) \leq 0 \end{array}$$

where f and $\mathbf{g}$ are convex, $\mathbf{h}$ is affine (linear plus a constant). The dual is given by

$$\text{(D)} \quad \text{Max}_{\mu \geq 0, \lambda} \phi(\lambda, \mu)$$

where $\phi(\lambda, \mu) = \text{Min}_{\mathbf{x}} [f(\mathbf{x}) + \mathbf{h}(\mathbf{x})\lambda + \mathbf{g}(\mathbf{x})\mu]$ with $\mu \geq 0$.

Theorem 8.3. *The following statements hold:*

- (i) *any local minimum point of (P) is a global minimum point;*
- (ii) *$-\phi(\lambda, \mu)$ with $\mu \geq 0$ is convex;*
- (iii) *If $\mathbf{x}^*$ is a regular point of the constraints and a global minimum point of (P) with corresponding value r^* and Lagrange multipliers $\mu^* \geq 0$ and λ^* , then (D) has a global maximum with respect to $\mu \geq 0$ and λ at μ^* and λ^* with corresponding value r^* .*

EXAMPLE 8.3. Find the dual of the problem

$$(P) \quad \begin{array}{ll} \text{Min} & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & A\mathbf{x} \geq \mathbf{b} \end{array}$$

SOLUTION. According to the definition, the dual function is

$$\begin{aligned} \phi(\mu) &= \text{Min}_{\mathbf{x}} \{ \mathbf{c}^T \mathbf{x} + \mu^T (\mathbf{b} - A\mathbf{x}) \} \quad (\text{with } \mu \geq 0) \\ &= \text{Min}_{\mathbf{x}} \{ (\mathbf{c}^T - \mu^T A)\mathbf{x} + \mu^T \mathbf{b} \}. \end{aligned}$$

Then

$$\phi(\mu) = \begin{cases} \mu^T \mathbf{b} & \text{for all } \mu \text{ such that } \mathbf{c}^T - \mu^T A = 0 \\ -\infty & \text{for other values } \mu. \end{cases}$$

Hence the dual

$$\begin{array}{ll} \text{Max} & \phi(\mu) \\ \text{s.t.} & \mu \geq 0 \end{array}$$

is

$$\begin{array}{ll} \text{Max} & \mu^T \mathbf{b} \\ \text{s.t.} & \mathbf{c}^T - \mu^T A = 0 \\ & \mu \geq 0, \end{array}$$

or

$$(D) \quad \begin{array}{ll} \text{Max} & \mu^T \mathbf{b} \\ \text{s.t.} & \mu^T A = \mathbf{c}^T \\ & \mu \geq 0. \end{array}$$

Similarly, we can establish the reverse direction. In summary,

$$\begin{array}{ll} \text{Min} & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & A\mathbf{x} \geq \mathbf{b} \end{array} \quad \Longleftrightarrow \quad \begin{array}{ll} \text{Max} & \mu^T \mathbf{b} \\ \text{s.t.} & \mu^T A = \mathbf{c}^T \\ & \mu \geq 0. \end{array}$$

EXAMPLE 8.4. Show that the linear system $A\mathbf{x} \geq \mathbf{b}$ has no solution iff there exists $\mathbf{y} \geq 0$ such that $\mathbf{y}^T A = \mathbf{0}$ and $\mathbf{y}^T \mathbf{b} > 0$.

Proof. Sufficiency Suppose to the contrary that there exists $\mathbf{x}$ such that $A\mathbf{x} \geq \mathbf{b}$. Then for the specified $\mathbf{y} \geq 0$, we have $\mathbf{y}^T (A\mathbf{x} - \mathbf{b}) \geq 0$, or $(\mathbf{y}^T A)\mathbf{x} \geq \mathbf{y}^T \mathbf{b}$ whose LHS = 0, while RHS > 0, a contradiction.

Necessity Suppose that for all $\mathbf{y} \geq 0$ with $\mathbf{y}^T A = \mathbf{0}$, we have $\mathbf{y}^T \mathbf{b} \leq 0$. Consider the problem

$$\begin{array}{ll}
 \text{(P)} & \begin{array}{ll} \text{Max} & \mathbf{y}^T \mathbf{b} \\ \text{s.t.} & \mathbf{y}^T A = \mathbf{0}^T \\ & \mathbf{y} \geq 0. \end{array}
 \end{array}$$

Then (P) achieves maximum at $\mathbf{y} = 0$. Note that the dual of (P) is

$$\begin{array}{ll}
 \text{(D)} & \begin{array}{ll} \text{Min} & \mathbf{0}^T \mathbf{x} \\ \text{s.t.} & A\mathbf{x} \geq \mathbf{b}. \end{array}
 \end{array}$$

By the duality theorem, 0 is the minimum value of (D). It follows that $A\mathbf{x} \geq \mathbf{b}$ has a solution, a contradiction. □